

Introduction to Interface

Operating condition

Operating voltage (12v / 5v / 3.3v)

Microcontroller / Interface voltage

5v tolerant

Level Translator

Exercise : measure digital pin output voltage

Digital Input Output

Exercise : 1 push button to turn on/off LED 2 counter to measure how many times the button is pushed

apakah perlu de-bouncing, how to?

Serial Port

Concepts :

port vs bus

hardware signal (RTS, etc)

software level – data format (bit level) : start/stop/bit rate/etc

Difference between RS232 (12v, DB-9 pin) and TTL (5v or other voltage, pin)

Exercise :

irim data ke PC

coba software serial

ADC / DAC

Concept :

Resolution (sampling speed, quantization)

Interface – Intermediate (Digital I/O)

I2C

Concepts : bus, speed, pins, configuration : hw/sw (pullup etc)

Limitation

Exercise : LCD display

OneWire

Concept

Limitation

Exercise : read DS18B20

SPI

Concept

Exercise : read / write W25Q31

RS485 bus

Concept : bus, serial with differential

Limitation

Exercise :

ubah komunikasi serial port dengan RS485
multiple device ?

Interface : Advanced

Modbus

Concept : protocol with RS485 bus

Communication

Modem

BLE

LoRa

Measurements
